
Update: Sheeraja's Feedback Items - All Core Tasks Completed

Devroop Kar (RIT Student) <dk7405@rit.edu>

Fri, Feb 13, 2026 at 10:56 AM

Reply-To: dk7405@rit.edu

To: pzg8794@rit.edu

Sounds good, thanks! You might want to look at this paper too - <https://ieeexplore.ieee.org/document/10621263>

Thanks and regards,
Devroop Kar

On Thu, Feb 12, 2026 at 4:40 PM Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:
Hi Devroop,

Perfect timing—I'll go through the comments and make the revisions this weekend. I'm also going to add the epsilon + NeuralUCB results, along with an additional comparison table for testbed runs using both their configuration and our run configuration (4k base frame, 2k step frame, 3–5 runs). This should give us a clearer comparison and more insight into which configuration works best for quantum routing.

Best,
Piter

On Thu, Feb 12, 2026 at 4:19 PM Devroop Kar (RIT Student) <dk7405@rit.edu> wrote:
Hi Piter,

Left some comments in the draft for you. Still going through it.

Thanks and regards,
Devroop Kar

On Sun, Feb 8, 2026 at 4:08AM Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:
Hi Dan,

I'm writing to share an update on the work completed since my last email. In addition to the previously completed items from Sheeraja's checklist, I've now finished the cross-testbed validation and results table revisions.

Previously Completed (Manuscript-side) items from Sheeraja's checklist

Definitions & technical clarifications

- Added explicit definitions for Tb, Fb, T, and Fc with consistent mathematical notation.
- Defined hop count and clarified intermediate repeater terminology.
- Added missing EXP3/NeuralUCB-related citations where referenced.
- Clarified variable usage and notation where ambiguity existed.

Figures & diagrams

- Corrected hop counts in the network topology diagram (Fig. 2) and aligned path labeling P1–P4.
- Updated the caption to explicitly state the hop definition used.

Tables & formatting

- Standardized column ordering across tables to keep metrics/scenarios consistent.
- Added legends/definition keys where they were missing (including the tables Sheeraja flagged, e.g., Tables 1, 3, 5–8).
- Cleaned Table 8 formatting to match the same column/legend conventions.

Content revisions

- Expanded the adversarial threat taxonomy (Sec. 3.3) using full sentences and explicit mechanics per regime.
- Added the uncertainty calculation explanation, including the epistemic uncertainty formulation and how it is used in the workflow.
- Revised allocator/model wording for uncertain regimes to remove ambiguous "may/might" phrasing where it weakened the claim.

Newly Completed (Cross-Testbed Validation & Results Tables)

Data validation & corrections

- Validated Tables 10 and 11 against all seven evaluation corpora (Papers 2, 7, 12 + CMABs, iCMABs, Hybrid, EXP3) — 0 data mismatches remaining.

Table 11 — Model Family Comparison (revised)

- Standardized filtering to the default allocator across all families. Previously, Hybrid aggregated across all four allocators while the other families only had Default, inflating Hybrid's denominator (300 vs 75 configs). All families now report over 75 identical configurations (5 scenarios × 3 scales × 5 experiments).
- Extended Table 11 with Default-allocator-only results from all three external testbeds (Papers 2, 7, 12), enabling direct cross-topology comparison against the internal Hybrid family.

Inter-Family Analysis (new/revised prose)

- iCPursuitNeuralUCB is the experiment winner across all six evaluation settings (internal corpus + 3 external testbeds), confirming topology-invariant dominance.
- Contextualized Paper 12's lower efficiency (~42%): the Oracle achieves 87% of the theoretical maximum reward and individual configurations reach up to 63% of Oracle, so the gap reflects bandit learnability difficulty on the 100-node/4-path fusion topology rather than a testbed fidelity ceiling.

Repo documentation

- Completed README contribution section and attribution updates.

Pending (non-manuscript / integration items that remain)

- Update the paper based on Professor Travis' feedback (once I get it).
- Integration of Sheeraja's UDRM / uncertainty-aware replay buffer logic into the framework.

All table values in the manuscript have been programmatically validated against the source CSVs with zero mismatches.

Note: *Should we set up a meeting to discuss about a quantum project proposal for my DSCI 601 class?*

Best,
Piter Garcia

On Sat, Feb 7, 2026 at 8:47 AM Daniel Krutz <dxkvse@rit.edu> wrote:

Hi Piter, I'm sorry for losing touch (the last few weeks have been nuts.) What are you working on now? Can you send me a 1-2 paragraph writeup?

Hi Dan,

No worries at all! Here's the quick reference:

WCCI 2026 (IJCNN) Submission

- Deadline: Saturday, January 31, 2026
- Primary Overleaf (working draft): <https://www.overleaf.com/project/68ea344896594f27b427ca8f>
- *(This is the active version with all current revisions—please review this one)*

Version Clarification

We currently maintain two Overleaf projects:

1. Main Working Version (link above)
— Contains all completed revisions from Sheeraja's feedback (definitions, figures, tables, threat taxonomy). This is the submission-ready draft.
2. Travis Review Version (focused on specific sections, such as research questions and results): <https://www.overleaf.com/read/ttmrwxvfqjfy38bfea>
3. We're awaiting his comments to then add additional changes into the main version.

Current Status

- All Sheeraja's manuscript feedback items completed (as detailed in my Jan 19 email) (Done)
- Paper 2 testbed integration full validation and jobs (as of last night—see Jan 28 email) (Done)
- Paper 7/12 testbed integration full validation and job (comparison tables coming by EOW) (validation Done, jobs running)

- Awaiting Professor Travis' feedback to integrate before any submission; btw, he just confirmed he is going to work on it now

I'll send a final pre-submission status update tomorrow with any remaining action items before the Saturday deadline.

Let me know if you need anything else!

Best,
Piter

On Tue, Jan 27, 2026 at 7:22AM Daniel Krutz <dxkvse@rit.edu> wrote:

Hi Piter, sorry for the late reply. When you don't hear from me for more than 1-2 days then ping me again. I believe that you're making a submission to WCCI. When is it due and what is the overleaf doc for this again? Sorry, there are a bunch of versions floating around and I want to be sure that I am looking at the right one.

Thanks
-Dan

On Jan 19, 2026, at 3:36AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi
Dan,

I'm writing to share an update on Sheeraja Rajakrishnan's feedback from our recent review meeting. I've completed the manuscript revisions that directly address her technical clarity and presentation items.

*Completed
(Manuscript-side) items from Sheeraja's checklist*

Definitions
& technical clarifications

- Added explicit definitions for Tb, Fb, T, and Fc with consistent mathematical notation.
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- Revised allocator/model wording for uncertain regimes to remove ambiguous “may/might” phrasing where it weakened the claim.

Pending (non-manuscript / integration items that remain)

- Testbed reruns for Papers 2/7/12 (for code/data validation).

- Comparison table in the paper once the testbed results are in.
- Update the paper based on Professor Travis' feedback (once I get it)
- Repo documentation updates (README contribution section + attribution).
- Integration of Sheeraja's UDRM / uncertainty-aware replay buffer logic into the framework.

All changes above directly address Sheeraja's technical feedback from the review session. The manuscript now has improved clarity in definitions, consistent table formatting, and expanded technical explanations as requested.

I'll send a separate email shortly about a possible submission pathway for WCCI 2026 and a plan for next steps based on my conversation with Professor Travis and Sheeraja.

Best,
Piter Garcia